

DSA 301, Project 3

Map Visualizations. NYC Taxi Data, April 5, 2016.

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Files in my zip

- DSA301_P3_34159_Gungor.html. My code and the maps.
- DSA301_P3_34159_Gungor.pdf. This file. I wrote it in Word and saved as PDF.
- Sketch photos for Task 1. Also inside the HTML file.

Task 1. Map Projections

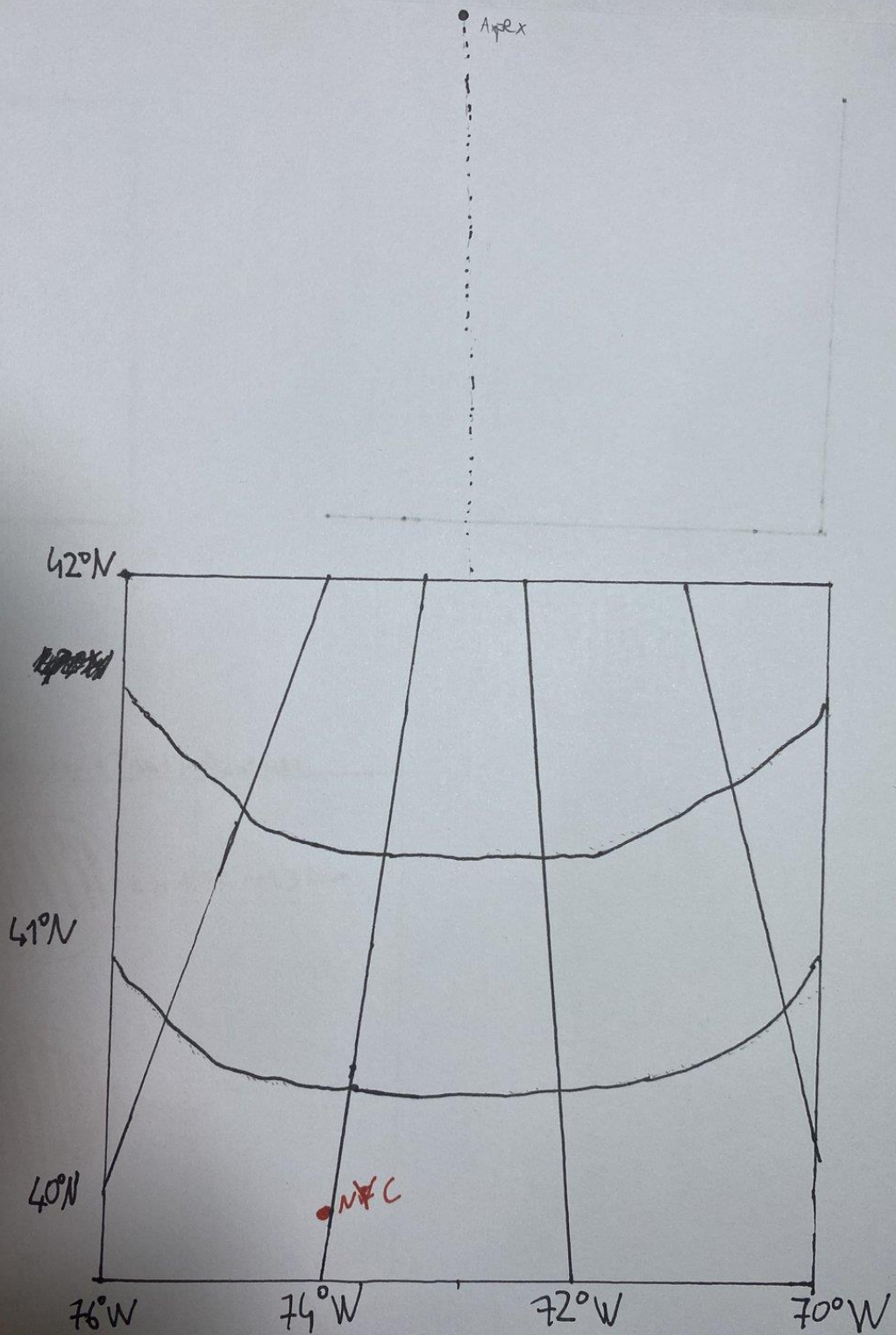
Paper and pen task. I compare Mercator and Albers Equal-Area Conic for the NYC area (70W to 76W, 40N to 42N).

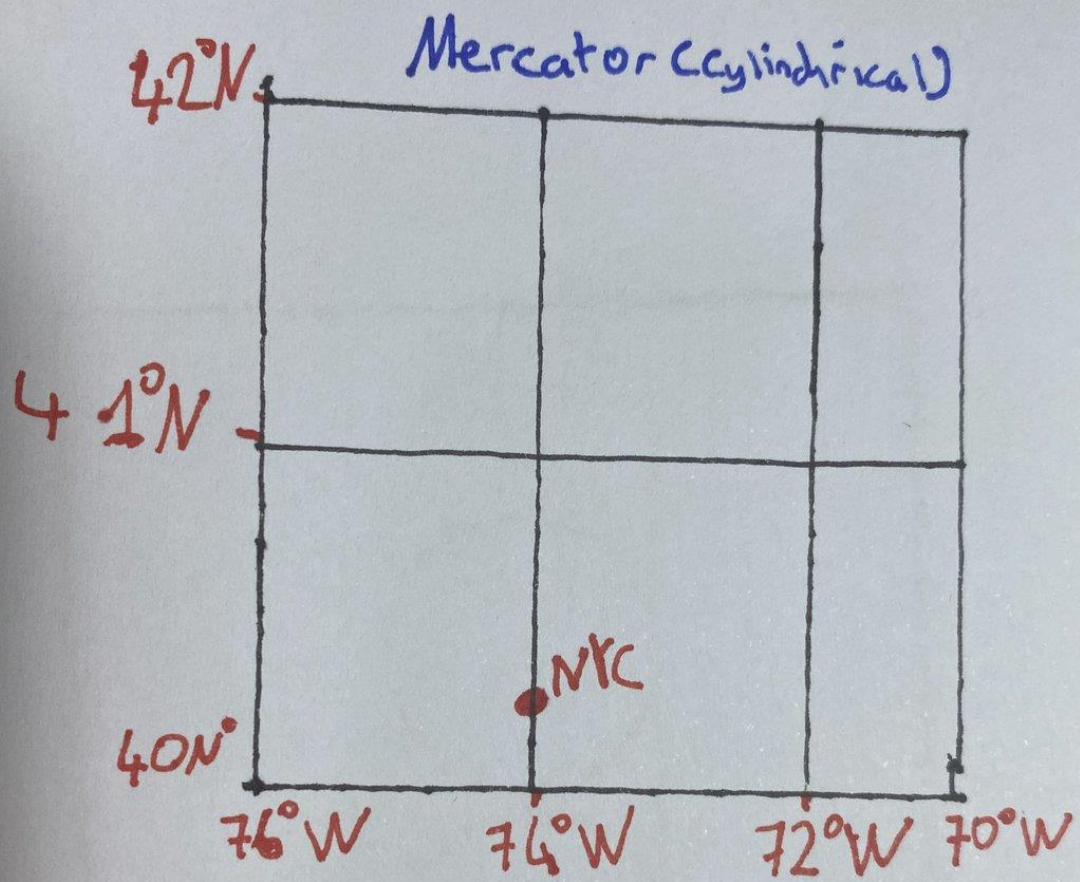
Part A. Graticule Sketch

I drew two boxes, about 10 cm each, on A4 paper.

Box 1 is Mercator. Mercator comes from a cylinder. When you unroll a cylinder you get a rectangle. So the meridians are straight vertical lines and the parallels are straight horizontal lines.

Box 2 is Albers Equal-Area Conic with standard parallels at 29.5N and 45.5N. It comes from a cone. When you unroll a cone you get a fan shape. The meridians meet at one point (the top of the cone, called the apex) and the parallels are curved arcs. I calculated the convergence angle as n is about $(\sin 29.5 + \sin 45.5) / 2$ which is about 0.564. So meridians 6 degrees apart converge by about 0.564 times 6 which is about 3.4 degrees at the northern edge of my box.



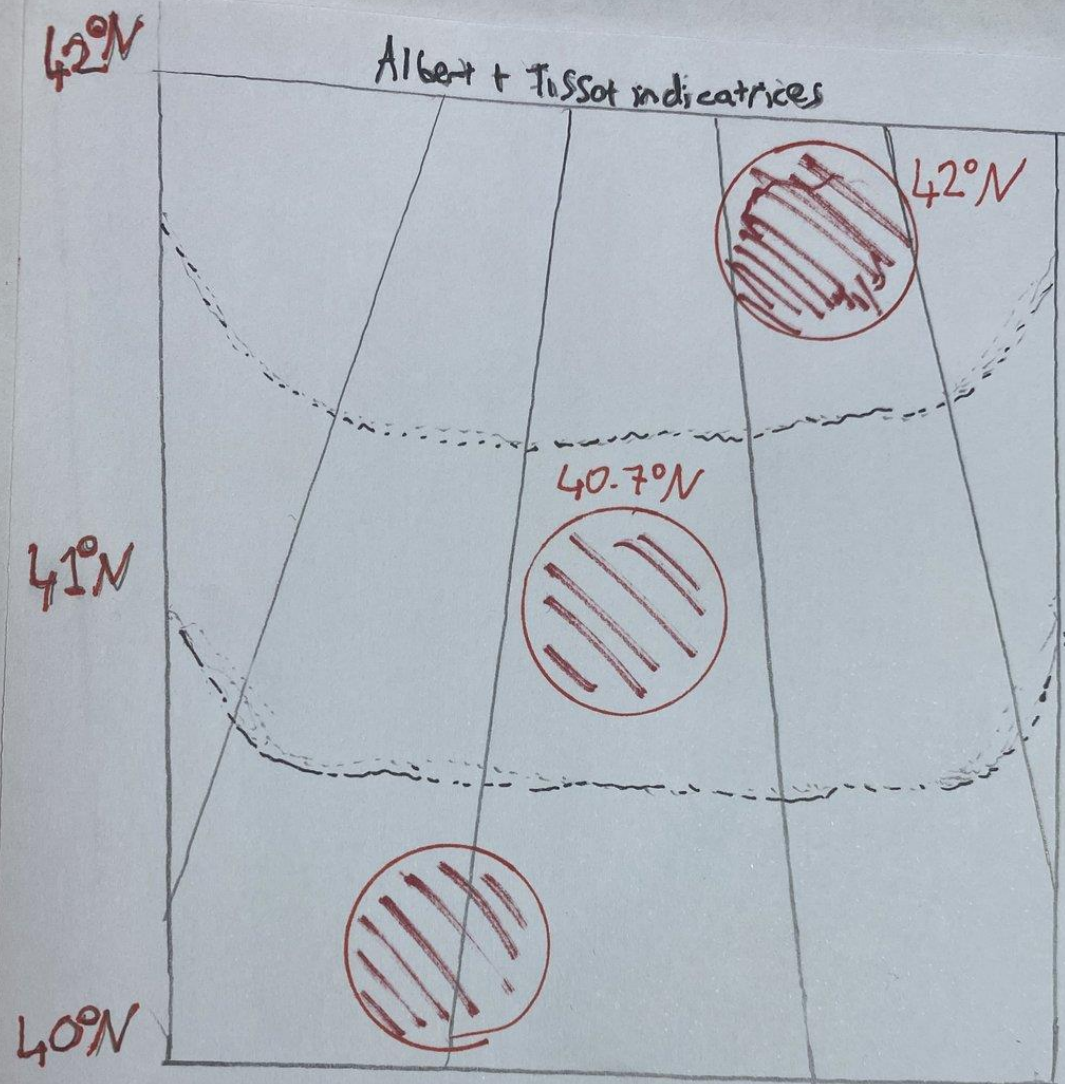


Part B. Tissot Indicatrices

I drew three small circles on each box. They are at 39N, 40.7N and 42N. The reference circle is 1 cm.

The formula is $k = 1 / \cos(\text{latitude}) = \sec(\text{latitude})$. The area is k times k . Big difference between the two projections: on Mercator the circles stay perfectly ROUND (because Mercator is conformal, it preserves local shapes) but they grow bigger at higher latitude. On Albers the circles are NOT round, they become slightly ELLIPTICAL (very hard to see because the scale error is less than 0.2%) but the AREA stays the same at every latitude. This is the round-vs-elliptical distinction: conformal keeps shape, equal-area keeps area, you cannot keep both.

- Mercator at 39N: $k = 1 / \cos(39)$ is about 1.29. Radius is 1.29 cm. Area = k squared is about 1.66.
- Mercator at 40.7N: k is about 1.31. Radius is 1.31 cm. Area = k squared is about 1.72 (this is the NYC area inflation factor from the project brief).
- Mercator at 42N: k is about 1.35. Radius is 1.35 cm. Area = k squared is about 1.82.
- Albers: circles are NOT round like on Mercator, they become slightly ELLIPTICAL because Albers preserves AREA but not shape. All three circles have the same area, but the shape is a little bit squeezed east-west as we move away from the two standard parallels (29.5N and 45.5N). Scale error is less than 0.1 percent at 40.7N and less than 0.2 percent at 42N, so the ellipse is almost round to the eye. I annotate "equal area" and mark the ellipse shape on the sketch.



39°N. Equal area

• Mercator + Tissot indicatrices



$k \approx 1.35$ $r = 1.35 \text{ cm}$



$k \approx 1.31$



$k \approx 1.29$

Part C. Written Analysis

Web map for NYC taxi data.

I pick Mercator. Web maps use tiles. Every zoom has a grid of equal square tiles, and every tile cuts into 4 equal smaller tiles at the next zoom. Only Mercator can do this. Also because Mercator is conformal, shapes stay consistent when you zoom in and out, so buildings and streets do not get distorted at any zoom. So the tile server needs Mercator. At NYC (about 40.7N) k is about 1.31. This is not perfect but for a city map it is fine. The heat layer is on the same tiles, so everything fits together.

Printed choropleth of pickups per km by borough.

For the print my answer changes. Choropleth uses rates, not totals, so the area must be correct. Then I need equal-area, for example Albers. If I use Mercator, some boroughs look too big and the density per km is wrong. The print is not interactive, so tile speed is not a problem. Albers is the right one here.

Part D. Projection Identification

Figure	Projection	Property	Clue from the graticule
Fig 3 (A)	Cylindrical, Mercator	Conformal (shape)	Meridians are straight and vertical. Parallels are straight and horizontal. The space between parallels grows near the top. The Tissot circles are round but bigger at higher latitude. This is Mercator.
Fig 4 (B)	Conic, Equal-Area	Equal-area	Meridians are straight lines that meet at one point above the map. Parallels are curved arcs. The Tissot shapes keep almost the same area. This is a conic equal-area map.

About the sketches: parts A and B are on A4 paper with a pen. I paste the photos above. I also put them inside the HTML file.

Task 2. Label Placement on a Live Map

Coded task. I put labels on the top 10 pickup cells. The map is in the HTML file.

Part A. Data Loading and Clustering

I open the file NYC-5April2016-TaxiTrips.xlsx. The coordinates are big integers. I divide longitude by 10 to the 15 and latitude by 10 to the 14. Then I filter to the NYC box (40.4 to 40.95N, -74.3 to -73.5W). I get 159 valid pickups. I put this data inside the HTML so the file works alone.

I make a grid of 0.01 by 0.01 degree cells. This is about 800 m by 800 m. I count the pickups in every cell. I pick the top 10. I draw a circle for every cell. Bigger count means bigger circle.

Part B. Label Implementation

For every marker I try 4 positions in order:

- Position 1: upper right. This is my first try.
- Position 2: lower right.
- Position 3: upper left.
- Position 4: lower left.

I use a bounding box test. If the label overlaps with another label I move to the next position. If all positions are busy I keep position 1. When the label is not in position 1 I draw a thin line from the marker to the label. I also put a white halo around the text. A halo helps to read the label on a busy map. Every label shows the rank and the trip count.

Part C. Written Reflection

C1. Zoom behaviour.

At zoom 14 the markers are far apart in pixels. So all 10 labels go to position 1 and the map looks clean.

At zoom 12 the Midtown markers are very close. Some labels go to position 2 or position 3 and they get a small leader line. My code runs the placement again on zoomend, so the labels are always placed again.

When I zoom out more, the labels start to touch. To fix this I can show only the top 3 or top 5 and hide the rest. I can also make the text smaller at low zoom, or make higher ranks bigger (size shows importance). I can also join very close cells into one label. These tricks keep the map readable.

C2. Figure and ground with the basemap.

The basemap has a lot of text already (street names, places). My labels and the basemap labels fight for attention. To make my labels read as the figure and the basemap as the ground (this is the Figure and Ground idea from Gestalt), I change these visual variables one by one: (1) halo thickness: a 2 px white stroke around every letter, so the label does not touch the grey basemap text; (2) size: my label is 12 px, a bit bigger than the OSM street text (10 to 11 px); (3) weight, I use font weight 700, and OSM labels are 400; (4) colour: my text is dark on a white halo, while basemap text is neutral grey, so the hue is different; (5) leader line, 1 px solid grey only when the label is not in position 1, so the line is quiet but still shows which cell the label belongs to. I also use a basemap with less colour so the ground stays quiet.

Task 3. Kernel Density Estimation

Coded task. I draw a smooth density map with Leaflet heat. The map is in the HTML file.

Part A. Full Dataset KDE

If I put a red dot for every pickup, the busy areas become one big blob. This is the "red dot fever" problem. Then the map mostly shows population, not taxi activity. A better way is binning or KDE. KDE puts a small smooth bump on every point and adds them. The result is a smooth surface.

I use all 159 pickups in one heat layer. I use a colour ramp with 5 stops. I also add a legend. I guess the scale from the data. The busiest cell has about 7 trips in one 0.01 by 0.01 cell (about 0.64 km squared). So it is about 11 trips per km squared. This is a sample day, a full day has more.

Part B. Bandwidth Sensitivity

Bin size (or radius) is very important. Too big is too coarse and I lose small patterns. Too small is too noisy and many cells are empty.

My slider goes from 5 to 60 px in steps of 5. When I move it, I remove the heat layer and add it again with the new radius. The slider also shows the value in metres. At zoom 12, 1 px is about 38 m.

My best value is about 25 to 30 px (about 950 to 1140 m). At this value I see Midtown and JFK as two different hotspots. Below 15 px the map is too noisy. Above 40 px Midtown and JFK become one big blob.

Part C. Temporal Slicing

I have three buttons for the time of day:

- Morning: 07 to 09.
- Midday: 12 to 14.
- Evening: 18 to 21.

There is also an "All day" button. When I click a button, the code filters the trips by hour and draws the heat layer again.

Morning (07 to 09). The map has one strong hotspot in Midtown Manhattan, a bit stronger near Penn Station and Grand Central where commuter trains arrive. People take a taxi from the station to the office. A smaller cluster is in the Financial District around Wall Street for the same reason. The Upper East Side also has a weak signal, maybe residents going to work. The outer areas (Brooklyn, Queens, Bronx) are mostly cold. JFK is very quiet because most flights arrive later in the day. The heat map shows a clear "commute into the centre" pattern, which fits the Red Dot Fever warning from the lecture: a raw dot map would mostly show where people live, but the heatmap shows where the activity is really concentrated.

Midday (12 to 14). The pattern is more spread out. Midtown is still the main hotspot but the intensity is a bit lower than morning. Upper East Side, Upper West Side and the Village also show warm cells, which looks like lunch and shopping trips. The Financial District is quieter than morning because workers are already inside their offices. JFK starts to become visible as midday flights land. The map is more balanced across the island of Manhattan. This is also why bandwidth sensitivity matters: if the radius is too big, these separate midday pockets merge into one blurry blob and the spatial spread is lost.

Evening (18 to 21). The map is the busiest window. Midtown stays hot and the heat extends south to Chelsea, Meatpacking and the Village (classic nightlife and restaurant areas). The west side near 34th to 42nd Street (Times Square area) is very bright, probably from shows and tourists. JFK is visible again because of evening arrivals and departures, and LaGuardia also gives a small warm cell. The outer boroughs are still cool but a bit warmer than morning. This temporal change is exactly the reason the lecture recommends filtering and re-running KDE on subsets: one "all day" map hides three very different stories.

Academic Honesty

For presentation only, I used a UI styling library (UI-UX Pro Max) for cosmetic help with the HTML page: the dark light theme colours, fonts, spacing, and card layout. It did not write any of the project logic. I also used an AI tool for grammar checks on my English writing, because my English is not perfect and I wanted the text to be easy to read. The AI only corrected small grammar mistakes. All the course work is my own: the coordinate decoding from the xlsx (divide by 10 to the 15 and 10 to the 14), the 0.01 degree grid clustering, the 4-position priority label placement with the bounding-box overlap test, the Leaflet heat layer configuration, the bandwidth slider, the temporal filter, and every written answer (Parts 1C, 1D, 2C, 3B, 3C). I read the lecture slides and the project brief myself and I chose how to solve every task.

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